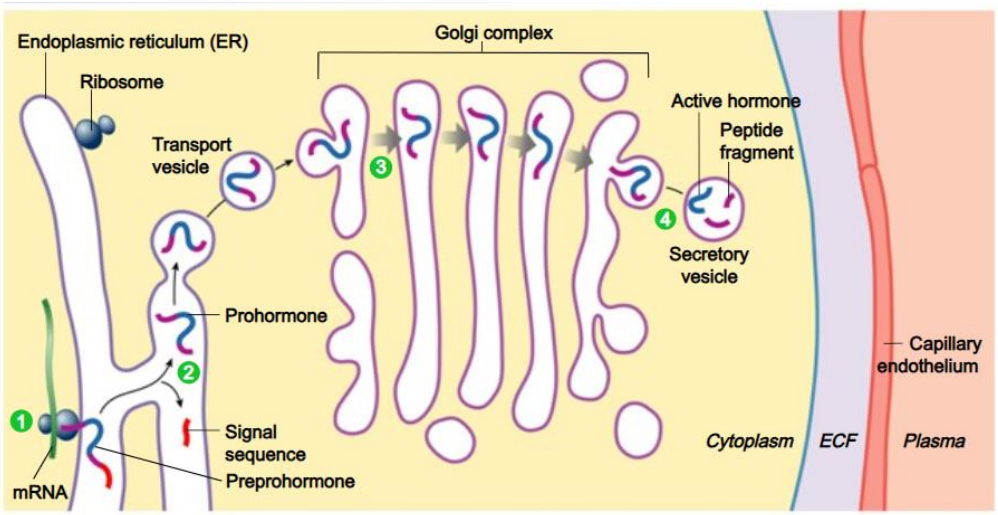
	311-211 Endocrine system and metabolism แนวข้อสอบ **MCQ** 1.ข้อสอบมี **1** ตอน **71** ข้อ 2.นักศึกษามีเวลาทำ **MCQ 2** ชั่วโมง 3.ข้อสอบเป็นภาษาอังกฤษทั้งหมด ***หมายเหตุ: ข้อสอบฉบับนี้เกิดจากความร่วมมือของคนในรุ่นและฝ่ายวิชาการ MED รุ่น 50

คะแนนเต็ม	MAX	MIN	MEAN	S.D.	จำนวนผู้สอบ
71.00	65.00	15.00	43.93	10.73	192

1. From the synthesis and secretion of hormone in the following picture.
*(ไม่ใช่ภาพจริงในข้อสอบ)



จากภาพเป็นการหลั่งของฮอร์โมนใดต่อไปนี้

A. Estrogen

B. Vitamin D

C. Aldosterone

D. Glucagon

E. Thyroid hormone

2. Which of following is a cellular mechanism of insulin in promoting GLUT4 translocation in skeletal muscle?

A. Activation of pip2

B. Activation of adenylyl cyclase

C. Activation of phospholipase C

D. Exchange of GDP to GTP in G alpha

E. Phosphorylation of tyrosine kinase

3. Action ของ Oxytocin ที่กระตุ้นให้ uterus หดตัว

A. Coupled to G_s to activate Adenylyl cyclase and using cAMP as second messenger

B. Coupled to G_i to inhibit Adenylyl cyclase and using cAMP as second messenger

C. Coupled to Guanylyl cyclase to inhibit phospholipase A_2 and using cGMP as second messenger

D. Coupled to G_q to activate the phospholipase C and using IP_3 as second messenger

E. Coupled to Tyrosine kinase using JAK-STAT signalling pathway.

4. -

5. A 46-year-old man reported that he has excessive thirst and has been drinking large quantities of water up to 10 liters per day for the past week. However, during physical examination he has poor skin elasticity and dry mucous membrane.
Which of the following structures is the most likely malfunction in this man?

A. Chief cells

B. Corticotroph

C. Somatotroph

D. Supraoptic nuclei

E. Paraventricular nuclei

6. ยกตัวอย่างฮอร์โมนที่สร้างจาก Basophil *เติมช้อยส์เพิ่มเอง (คิดเอง)

A. Growth hormone

B. Prolactin

C. Aldosterone

D. Thyroid-stimulating hormone

E. Antidiuretic hormone

7. จากการ Overexpression of ADH ส่งผลให้เกิดอะไรบ้าง

A. Polyuria

B. Polydipsia

C. Dehydration

D. Polyphagia

E. hyponatremia

8. Which drug should be given to the patient with growth retardation?

A. Somatropin

B. Lanreotide

C. Mecasermin

D. Pegvigsomant

E. Ganirelix

9. A young women มี puffy skin, weight gain, TSH concentration low TRH markly increase when she is given. What is the following pathology associated with this patient?

A. Hyperthyroidism due to a thyroid tumor

B. Hyperthyroidism due to abnormality in the hypothalamus

C. Hypothyroidism due to abnormality in the hypothalamus

D. Hypothyroidism due to abnormality in the Thyroid

E. Hypothyroidism due to abnormality in the Pituitary

10. Which of the following cause of goiter in Grave's disease?

A. Iodine deficiency

B. High level plasma TRH

C. High level plasma TSH

D. High level plasma TSH receptor antibodies

E. Increase of immune TSH receptor antibody*** มันเหมือนกับข้อ D. หรือเปล่า?

11. A 20-year-old woman presents with neck pain for 2 days. PE: P 110/min, mild enlargement of thyroid gland with marked tenderness. Lab: free T4 1.90 ng/dL (normal 0.7-1.7), TSH 0.05 mIU/L (normal 0.25-4.00). Which of the following is the thyroid disease in this patient?

A. Grave’s disease

B. Subacute thyroiditis

C. Papillary carcinoma

D. Follicular carcinoma

E. Hashimoto's thyroiditis

12. A 35-year-old man presents with a solitary thyroid nodule in right lobe. He unwanted surgical resection. The histologic sections showed a completely encapsulated mass, no evidence of cell expanding beyond capsule and no follicle cell in vessel. The remaining thyroid follicles look normal. What is the most likely diagnosis?

A. Follicular adenoma

B. Follicular carcinoma

C. Follicular variant in papillary carcinoma

D. Follicular lesion of undetermined significance

E. Nodular goiter

13. -

14. Mechanism หลักของ iodine คืออะไร

A. Inhibit thyroglobulin proteolysis

B. Block Wolff-Chaikoff effect

C. Inhibit thyroid peroxidase

D. Block coupling of iodotyrosines

E. Destroy thyroid parenchyma

15. Which of the following is a common adverse effect of radioactive iodine?

A. Acneiform Rash

B. Fever

C. Tremer

D. Hypothyroidism

E. Severe hepatitis

16. มารดาอายุครรภ์ 4 เดือนควรใช้ยา Antithyroid drug ชนิดใดต่อไปนี้

A. PTU

B. MMI

C. Iodides

D. Radioactive iodine

E. Propranolol

17. A patient presented with pheochromocytoma. She was heart palpitation, hypertension and hyperglycemia. To determine the cause of excessive of Catecholamines, which of the following of clinical biomarker can be measured?

A. Tyrosine

B. C-peptide

C. Cortisone

D. Chromogranin

E. Acetylcholine

18. (คล้ายข้อ 13 ปี 49 แต่เปลี่ยนยีน จำยีนไม่ได้ TT) A patient with incomplete mutations of the starD1 gene, show low level of STAR protein expression. This mutation led to elevated ACTH, adrenal hypertrophy, and engorgement of the adrenal cell with lipid. Which of the following is the symptoms of the patient?

A. Salt waste

B. Hypokalemia

C. Hypertension

D. Hyperglycemia

E. Metabolic alkalosis

19. –

20. –

21. –

22. ให้อาการมา thyroid hormone TSH ปกติทุกอย่าง แต่ testosterone กับ ACTH เพิ่ม ถามว่าผู้ป่วยมีอาการอย่างไร

23. –

24. ผลแลปได้ Hypokalemia ต้องตรวจอะไรเป็นอันดับแรก

A. Plasma ACTH

B. Urinary free cortisol

C. 11 PM serum cortisol

D. Aldosterone to rennin ratio

E. Urine vanillylmandelic acid

25. เป็น Addison disease แล้ว ACTH ลดลง cortisol ลด test อะไร แยกระหว่าง pituitary กับ hypothalamus defects

A. ACTH stimulation

B. CRH stimulation

C. Dexamethasone suppression

D. Serum cortisol

E. Plasma ACTH

26. –

27. ยาที่ใช้รักษา Acromegaly

28. Patient with IGF-1 deficiency was treated with mecasermin. The most common ADR?

A. Hypokalemia

B. Hypoglycemia

C. Capal tunnel syndrome

D. Retinopathy

E. Hypotention

29. –

30. คนไข้ Cushing's syndrome มีอาการ hypertension ให้ยาใดต่อไปนี้

31. ผู้ป่วยกิน prednisolone มา 5 ปี ตอนนี้มีเบาหวาน เบาหวานที่เกิดจากการใช้ยาเกิดได้อย่างไร

32. คนไข้เป็น DM1 insulin deficiency ถ้าขาด insulin จะทำให้เกิดอะไรบ้าง

33. Which of the following conditions increases plasma insulin levels?

A. Hypoglycemia

B. After intravenous administration of somatostatin

C. Following vigorous exercise

D. Following a carbohydrate-rich meal

E. During a surgical procedure

34. What is the following factors increase insulin secretion in beta cell of pancreas?

A. Decrease cAMP

B. Decrease ---

C. Increase oxidative phosphorylation

D. Increase K+ efflux via ATP sensitive K+ channel

E. Increase Ca2+ efflux

35. A 47-year-old woman อาการ fatigue confusion blurred vision หิวมากตอนเช้า กินข้าวแล้ว 20-30 นาทีหาย แพทย์สงสัย hypoglycemia จาก insulinoma (tumor หลั่ง insulin) ถามว่าข้อไหนตรงกับอาการ

A. Fructose 2,6 biphosphate เพิ่ม

B. cAMP dependent ... เพิ่ม

C. Glycogen synthase b เพิ่ม

D. Glycogen synthase dephospholyrase ถูกยับยั้ง

E. Affinity ของ fructose 1,6 biphosphatase กับ fructose 1,6 biphosphate เพิ่ม

36. A 45-years-old male with type 2 diabetes has been prescribed sulfonylurea for blood glucose control. Which of the following choices is the effect of this drug on pancreatic β cell?

A. ATP-gate K^+ channel is activated

B. Pancreatic β cell membrane depolarization

C. Insulin granules are secreted by voltage gated channel

D. Decreased intracellular concentration of Ca^{2+}

E. Voltage-dependent Ca^{2+} channel is inhibited

37. A 38-year-old woman with type 2 DM developed severe lactic acidosis after receiving after receiving anti-diabetic drugs. Which of the following drug is the likely cause?

A. Metformin

B. Acarbose

C. Repaglinide

D. Regular insulin

E. Pioglitazone

38. Which of the following is a main mechanism of glyburide glibenclamide lower blood glucose level?

A. Increase insulin synthesis

B. Promote glucose uptake by muscle, liver, and adipose tissue

C. Enhance renal excretion of glucose

D. Decrease insulin resistance by lowering body weight

E. Release insulin from the pancreas.

39. Which of the following pairs of oral antidiabetic agent work through complementary mechanism of action and therefor may target both primary pathophysiologic defect in type2 DM

A. Glimepiride and nateglinide

B. Metformin and canagliflozin

C. Acarbose and miglitol

D. Somatropin and dopamine

E. Denosumab and calcitonin

40. หากผลตรวจ Hemoglobin typing ของคนไข้ได้เป็น EE ต้องใช้อะไรในการติดตามระดับน้ำตาลของคนไข้?

A. Fructosamine

B. HbA1C

C. Urine

D. Oral glucose tolerance test

E. Free form albumin

41. FPG 120 mg/dl และ 75g OGTT 135 mg/dl รายงานผลอย่างไร?

A. DM

B. Normal

C. Impaired fasting glucose

D. Impaired glucose tolerance

E. Hyperthyroidism

42. Which of the following processes would be most affected in a person with LCAT deficiency?

A. Binding of LDL to its receptor

B. Conversion of discoidal to spherical HDL particle

C. Conversion of Chylomicron-to-Chylomicron remnant

D. Transfers of cholesterol from LDL to peripheral tissues

E. Exchange of cholesterol ester and triglyceride between HDL and VLDL

43. A man has xanthomas over the knees, the wrists, and elbows. His serum LDL cholesterol is 800 (normal<200) mg/dL with HDL cholesterol of 40 mg/dL. He was diagnosed as having myocardial infarction at age 15. What is the mechanism of his dyslipidemia?

A. Decreased chylomicron synthesis

B. Increased hepatic lipid synthesis

C. Decreased lipid excretion

D. Increased hormone sensitive lipase activity

E. Decreased receptor-mediated endocytosis of LDL

44. Why does deficiency of lipoprotein lipase cause hypertriglyceridemia?

A. Triglyceride in IDL is not hydrolyzed

B. Triglyceride in LDL is not hydrolyzed

C. Triglyceride in VLDL is not hydrolyzed

D. Triglyceride in adipose tissue is not hydrolyzed

E. Less nascent chylomicron can be converted to mature chylomicron

45. What can be first visible in atherosclerosis plaques?

A. Amyloid fibril

B. Calcification

C. Fatty streak

D. Hemorrhage

E. Thrombus

46. ผู้ป่วยเป็น DM smoke cigarette 1 pack a day เป็น Myocardial infarction ถามว่าอะไรที่เสียหายเป็นอย่างแรก?

A. Endothelial

B. Subendothelial

C. Macrophage

D. Red blood cell

E. Lung

47. Increase HDLs level decrease risk altherosis which of the following anti-atherogenic action HDLs?

A. Cholesterol efflux from tissue

B. Activate LDL เข้าตับ

C. เปลี่ยน Cholesterol ให้เป็น bile acid

D. ทำลาย Cholesterol ในตับ

E. ลดจำนวน LDL receptor

48. 15-years-old patient มีอาการ hypercholesterolemia ได้รับการรักษาด้วย statin แล้วระดับ LDL ยังไม่ลด เมื่อตรวจยีนพบว่ามี overexpression of PCSK9 gene ควรใช้ยาอะไรในการรักษา?

A. Alirocumab

B. Ezetimibe

C. Simvastatin

D. Cholestyramine

E. Denosumab

49. A 55-year-old patient with Cushing syndrome has persistence dyslipidemia. He is being treated for Cushing syndromes with drug which potent CYP3A4 inhibitor. Which of following lipid lowering drug should be prescribed in this patient?

A. Niacin

B. Pravastatin

C. Simvastatin

D. Atorvastatin

E. Evolocumab

50. which of the following drugs increase LDL receptor production by decrease delivery chylomicron remnant to liver

A. Fenofibrate

B. Niacin

C. Ezetimibe

D. Colestipol

E. Atorvastatin

51. ให้ระดับ Tc 280 mg/dL, TG 150 mg/dL, LDL-c 220 mg/dL, HDL-c 30 mg/dL ถามว่าปริมาณ non-HDL cholesterol เป็นเท่าไหร่? (โจทย์ถามประมาณนี้ แต่แต่งตัวเลขขึ้นมา)

A. 280 mg/dL

B. 220 mg/dL

C. 250 mg/dL

D. 120 mg/dL

E. 180 mg/dL

52. Which of the following have highest risk of sd-LDL?

A. HDL-c 45, Total cholesterol 190

B. HDL-c 40, Total cholesterol 160

C. HDL-c 50, Total cholesterol 140

D. HDL-c 45, Total cholesterol 160

E. HDL-c 160, Total cholesterol 40

53. A 58-year-old man had a thyroid cancer removal. Six months later, he had a regular medical check-up. The doctor mentioned that he had an abnormal calcium level (6.8 mmol/L). Which of the following sign/symptoms might be observed?

A. Dehydration

B. Tetany

C. Osteoporosis

D. Kidney stones

E. Constipation

54. A 55-year-old man กิน thiazide diuretics เพื่อลด hypertension ต่อมาเกิด constipation, fatigue, plasma Ca^{2+} 12.5 mg/dL โจทย์ถามว่า Thiazide diuretics มี MOA อย่างไร?

A. ลด TRPV6 activity

B. ลด Intracellular Na^+

C. ลด Na^+ - K^+ - $2Cl^-$ cotransporter activity

D. เพิ่ม Na^+ - Cl^- cotransporter activity

E. เพิ่ม Ila Na^+/P_i cotransporter expression

55. Which of the following conditions cause decrease Ca^{2+} reabsorption in the kidney?

A. Hypocalcemia

B. Metabolic alkalosis

C. Hyperparathyroidism

D. Loop diuretic

E. Vitamin D supplement

56. What is upregulate in menopause female led to osteoporosis?

A. RUNX2

B. RANKL

C. OPG

D. BMP2

E. Alkaline phosphate

57. OPG ทำอะไรกับ RANKL?

A. Interact

B. Dimer

C. Degrade

D. Activate

E. Transcript

58. Drug of choices for treat osteoporosis in postmenopausal women at high risk of fractures?

A. Alendronate

B. Teriparatide

C. Denosumab

D. Calcitonin

E. Raloxifene

59. Which of the following drug should be prescribed as the adjunctive therapy for postmenopausal osteoporotic?

A. Raloxifene + Vitamin D

B. Calcium + Vitamin D

C. Magnesium + Vitamin C

D. Zinc + Vitamin C

E. Estrogen + Vitamin D

60. อาการเป็น Cushing disease ถามว่า Hormone อะไรเป็นตัวการ?

A. Cortisol

B. ACTH

C. Aldosterone

D. Growth hormone

E. Prolactin

61. Female patient presents with second amenorrhea with galactorrhea. Which adenoma should be found in this patient?

A. Cortical adenoma

B. Prolactinoma

C. Lipoma

D. Papillary carcinoma

62. What is the diagnosis in a patient who has hyperglycemia from total pancreatectomy and requires insulin injection?

A. Type 1 diabetes

B. Type 2 diabetes

C. Other specific type of diabetes

D. Diabetes of undetermined cause

E. Insulin dependent mellitus

63. 58-year-old-female presents with visual disturbance and headache. Magnetic Resonance Imaging (MRI) shows tumor in Sella turcica compressing optic chiasma. What is the embryonic structure that gives rise of Sella turcica?

A. Rathke's pouch

B. Neural crest cells

C. Notochord

D. Neural tube

E. Mesoderm

64. Which of the following mechanisms involved in glucose homeostasis during fed state?

A. Increase hepatic glycogenolysis

B. Increase urine excretion of glucose

C. Increase hepatic glucose production

D. Increase uptake glucose in brain

E. Increase uptake glucose in muscle

65. -

66. Which of Lipoprotein particles involved in providing energy for the cell?

A. IDL

B. LDL

C. HDL

D. VLDL

E. CM

67. -

68. Which of the following lipoprotein particles increased when ApoE abnormal?

A. IDL

B. DHL

C. LDL

D. VLDL

E. Chylomicron

69. -

70. -

71. ผู้ชายอายุ 20 ปี x-ray กระดูดขาพบ loss สักอย่าง ประสบอุบัติเหตุ รถชน 2 ปี ต้องนอนเฉยๆ ผลฟิมล์ที่เห็นมาจากอะไร?

A. Osteoclast

B. Osteoblast

C. Osteocyte

D. Adipocyte

E. Myoblast

- “*GOOD LUCK*” -